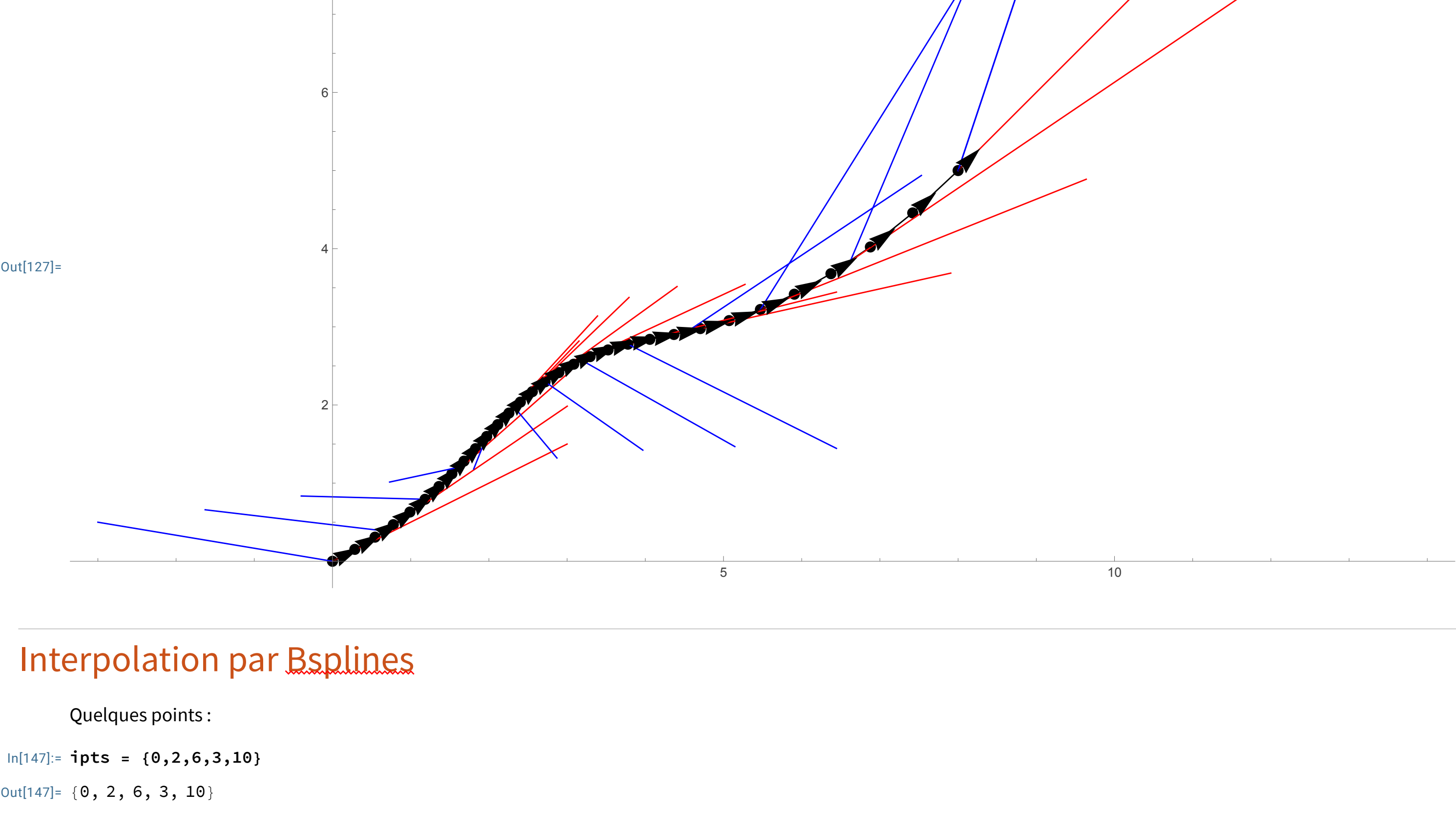
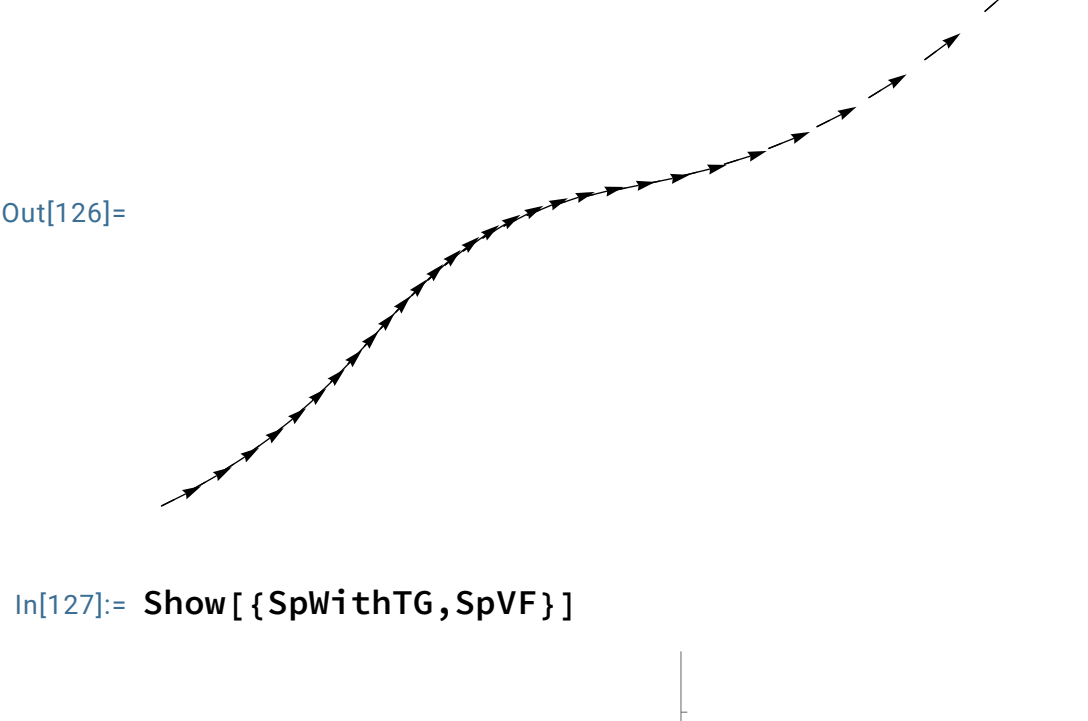


Champs des vecteurs vitesse

General: VectorFieldPlots is now obsolete. The legacy version being loaded may conflict with current functionality. See the Compatibility Guide for updating information.

```
In[123]:= SpVF = ListVectorFieldPlot[Table[[CSp[u],CSp'[u]],
{u,0,3,0.1}],
ScaleFunction->(.4k)]
```



Interpolation par Bsplines

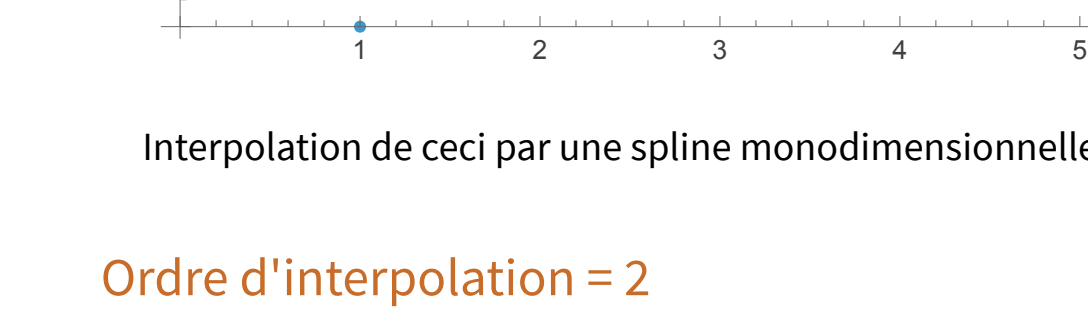
Quelques points:

```
In[147]:= ipts = {0,2,6,3,10}
```

```
Out[147]> {0, 2, 6, 3, 10}
```

Les points en (i,pts[i])

```
In[148]:= ListPlot[ipts,
Prolog -> {PointSize[0.02]}]
```



Interpolation de ceci par une spline monodimensionnelle et naturelle. Donc interpolation par une splines de 4 arcs et d'ordre k = (2,3,4).

Ordre d'interpolation = 2

```
In[157]:= knots = {-1,0,1,2,3,4,5}
```

```
Out[157]> {-1, 0, 1, 2, 3, 4, 5}
```

```
In[158]:= k = 2
```

```
Out[158]> 2
```

```
In[159]:= pars = Table[i-1,{i,1,5}]
```

```
Out[159]> {0, 1, 2, 3, 4}
```

```
In[160]:= Bs[1,k,knots][pars[[1]]]
```

```
Out[160]> 1
```

```
In[161]:= Table[Bs[i,k,knots][u],{i,1,5},{u,0,4}]
```

```
Out[161]> {{1, 0, 0, 0, 0}, {0, 1, 0, 0, 0}, {0, 0, 1, 0, 0}, {0, 0, 0, 1, 0}, {0, 0, 0, 0, 1}}
```

```
In[162]:= MatrixForm[%]
```

```
Out[162]> MatrixForm[
1 0 0 0 0
0 1 0 0 0
0 0 1 0 0
0 0 0 1 0
0 0 0 0 1]
```

```
In[163]:= unknown = Table[Q[i],{i,1,5}]
```

```
Out[163]> {Q[1], Q[2], Q[3], Q[4], Q[5]}
```

```
In[176]:= seq = Table[Sp[2,knots,unknown][pars[[i]]] == ipts[[i]],
{1,1,Length[ipts]}]
```

```
Out[176]> {Q[1] == 0, Q[2] == 2, Q[3] == 6, Q[4] == 3, Q[5] == 10}
```

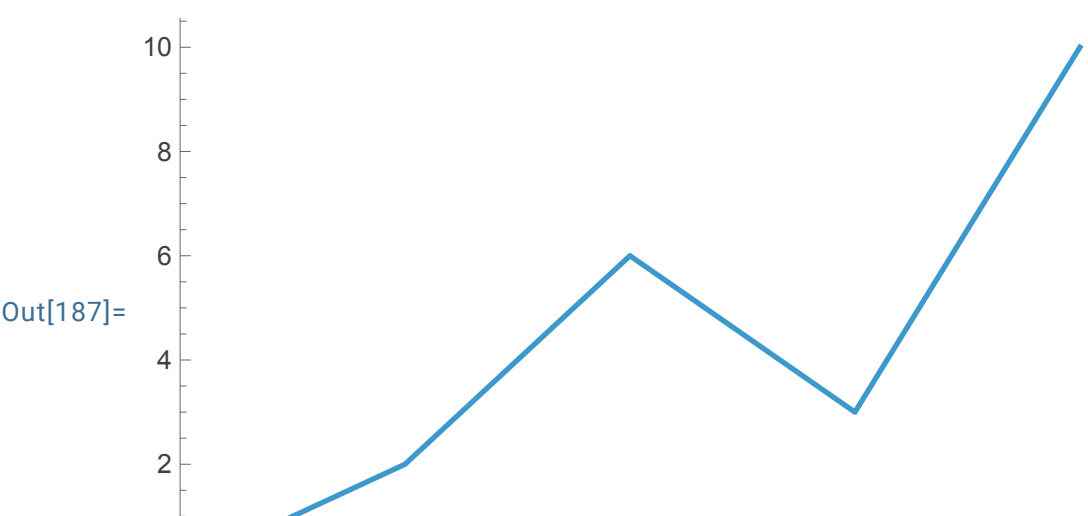
```
In[177]:= sr = Solve[seq, unknown]
```

```
Out[177]> {{Q[1] -> 0, Q[2] -> 2, Q[3] -> 6, Q[4] -> 3, Q[5] -> 10}}
```

```
In[179]:= sols = Map[Sp[2,knots,unknown /. #][u]&, sr]
```

```
Out[179]> {2 (dz[u, 1] - Bs[2, {-1, 0, 1, 2, 3, 4, 5}][u] - dz[2 - u, 1] - Bs[3, {-1, 0, 1, 2, 3, 4, 5}][u]) +
6 (dz[u, 1] - Bs[3, {-1, 0, 1, 2, 3, 4, 5}][u] + dz[3 - u, 1] - Bs[4, {-1, 0, 1, 2, 3, 4, 5}][u]) +
3 (dz[-2 - u, 1] - Bs[4, {-1, 0, 1, 2, 3, 4, 5}][u] + dz[4 - u, 1] - Bs[5, {-1, 0, 1, 2, 3, 4, 5}][u]) +
10 (dz[-3 - u, 1] - Bs[5, {-1, 0, 1, 2, 3, 4, 5}][u] + dz[5 - u, 1] - Bs[6, {-1, 0, 1, 2, 3, 4, 5}][u])}
```

```
In[187]:= Plot[sols,{u,0,4}]
```



Interpolation a l'ordre 3

```
In[188]:= ipts
```

```
Out[188]> {0, 2, 6, 3, 10}
```

```
In[189]:= k = 3
```

```
Out[189]> 3
```

```
In[190]:= knots = {-2,-1,0,1,2,3,4,5,6}
```

```
Out[190]> {-2, -1, 0, 1, 2, 3, 4, 5, 6}
```

```
In[191]:= unknown = Table[Q[i], {1,1,Length[knots]-k}]
```

```
Out[191]> {Q[1], Q[2], Q[3], Q[4], Q[5], Q[6]}
```

```
In[192]:= pars
```

```
Out[192]> {0, 1, 2, 3, 4}
```

```
In[193]:= interpolsPtsEqs =
Table[Sp[k,knots,unknown][pars[[i]]] == ipts[[i]],
{1,1,Length[ipts]}]
```

```
Out[193]> {Q[1] + Q[2] == 0, Q[2] + Q[3] == 2, Q[3] + Q[4] == 6, Q[4] + Q[5] == 3, Q[5] + Q[6] == 10}
```

5 equations pour 6 inconnues, manque une condition initiale:

vitesse nulle au depart

```
In[194]:= condsSupp = Sp[k,knots,unknown]'[0] == 0
```

```
Out[194]> 2 dz[-Q[1] + Q[2], 2] == 0
```

```
In[195]:= eqnsToSolve = Append[interpolsPtsEqs, condsSupp]
```

```
Out[195]> {Q[1] + Q[2] == 0, Q[2] + Q[3] == 2, Q[3] + Q[4] == 6, Q[4] + Q[5] == 3, Q[5] + Q[6] == 10, 2 dz[-Q[1] + Q[2], 2] == 0}
```

```
In[196]:= solutions = Solve[eqnsToSolve,unknown]
```

Solve: Inverse functions are being used by Solve, so some solutions may not be found; use Reduce for complete solution information

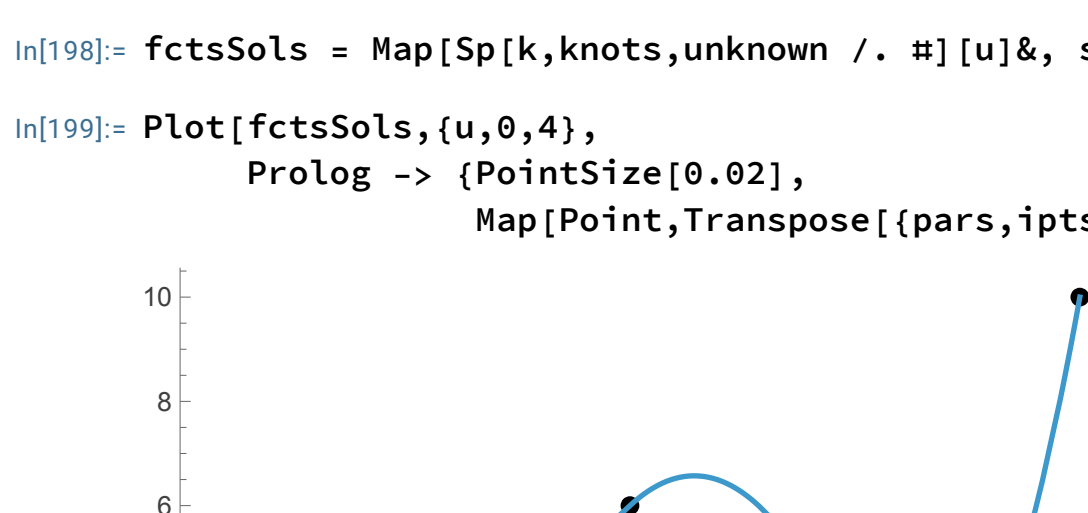
```
Out[196]> {{Q[1] -> 0, Q[2] -> 0, Q[3] -> 4, Q[4] -> 8, Q[5] -> -2, Q[6] -> 22}}
```

```
In[197]:= Transpose[{pars,ipts}]
```

```
Out[197]> {{0, 0}, {1, 2}, {2, 6}, {3, 3}, {4, 10}}
```

```
In[198]:= fctsSols = Map[Sp[k,knots,unknown /. #][u]&, solutions];
```

```
In[199]:= Plot[fctsSols,{u,0,4},
Prolog -> {PointSize[0.02],
Map[Point,Transpose[{pars,ipts}]]]}
```



acceleration nulle au depart

```
In[230]:= dz[x_, y_?NonZeroQ] := x/y
```

```
In[229]:= dz[1, 4]
```

```
Out[229]> 1/4
```

```
In[216]:= condsSupp = Derivative[2][Sp[k,knots,unknown]][0] == 0
```

```
Out[216]> Q[1] - 2 Q[2] + Q[3] == 0
```

```
In[217]:= eqnsToSolve = Append[interpolsPtsEqs, condsSupp]
```

```
Out[217]> {Q[1] + Q[2] == 0, Q[2] + Q[3] == 2, Q[3] + Q[4] == 6, Q[4] + Q[5] == 3, Q[5] + Q[6] == 10, Q[1] - 2 Q[2] + Q[3] == 0}
```

```
In[218]:= solutions = Solve[eqnsToSolve,unknown]
```

```
Out[218]> {{Q[1] -> -1, Q[2] -> 1, Q[3] -> 3, Q[4] -> 9, Q[5] -> -3, Q[6] -> 23}}
```

```
In[203]:= fctsSols = Map[Sp[k,knots,unknown /. #][u]&, solutions];
```

ReplaceAll: {Q[1], Q[2], Q[3], Q[4], Q[5], Q[6]} is neither a list of replacement rules nor a valid dispatch table, and so cannot be used for replacing.

ReplaceAll: {Q[1], Q[2], Q[3], Q[4], Q[5], Q[6]} is neither a list of replacement rules nor a valid dispatch table, and so cannot be used for replacing.

ReplaceAll: {Q[1], Q[2], Q[3], Q[4], Q[5], Q[6]} is neither a list of replacement rules nor a valid dispatch table, and so cannot be used for replacing.

Solve: Sp[3, {-2, -1, 0, 1, 2, 3, 4, 5, 6}] is not a quantified system of equations and inequalities.

Solve: Sp[3, {-2, -1, 0, 1, 2, 3, 4, 5, 6}] is not a quantified system of equations and inequalities.

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